



Recommended Minimum Requirements for PESALINK Technology Infrastructure, Security and Operations

Version 2.0

30th October, 2020

Requirements for Upgraded version of PesaLink (To Go Live by Q1, 2021)

Area	Recommended Minimum Requirements
General	- Middleware with 24x7x365 core instant payments processing capability supported by an integrated platform for straight-through processing (STP) - Payments and accounts verification/validation, extended with tools for anti-fraud monitoring and transaction status check - Immediate feedback. For example, an immediate reject is required in case funds are not available or the originating/beneficiary account does not exist or dormant - Account verification (Hakikisha) to enable customers validate recipient account name before committing financial transaction - Transaction failure monitoring and exception handling
Network	- 2 Links (primary and secondary) for high availability with 2Mbps capacity - Cisco 4300& Above Router - Next Generation Firewall with Intrusion Prevention System (IPS)
Application	- Core Banking Middleware (API Gateway or/and ESB) - for the financial transaction message exchange. Banks to separate interface for both inbound and outbound messages to ensure better throughput and processing performance - Incoming and outgoing payments messages format based on ISO20022 (and incoming messages into Bank's own format where required) - Set of payment API endpoints that will developed according to PesaLink Interface specification for communication with the PesaLink Switch - XML format Validation (ISO 20022 message tags and elements check) - Duplicate Validation before sending transactions to PesaLink - Validating the correctness of the payment instructions and determines payment processing flow - Replying authentication and authorization requests sent from PesaLink
Technology Infrastructure	Hardware (Product variants should comply with minimum technology specifications) • Consideration for dedicated infrastructures for PesaLink Interface integration and transaction processing with high availability • Minimum of G7/G8 Server or any other product variants with minimum of 16 CPU processors • Minimum of 500 GB HDD (or SSD based on internal and capability of the bank/institution) • 32 GB RAM or more Operating System (Windows OS) • 64-bit Operating Systems (MS Server 2019 at the minimum) Registry Settings • Time Wait-Delay should be set at 10secs • MaxPort set at 65534 (Linux OS) • 64-bit Operating Systems (Linux/Unix Sever with Commercial Support) Registry Settings

	• Time Wait-Delay should be set at 10secs • Max port set to 65534 Policy on regular patching and anti-virus update Database • Policy on regular purging and fine-tuning of database
Risk and Security	- Public- Key Cryptography - End-to-End encryption of transaction message during processing - Digital Certificate/ Security Validation - Business rules validation for incoming transaction (validation against core banking system for account name and status of account) before crediting customer and responding to PesaLink - Business rules validation for outgoing transaction (account balance, validate rules and authentication before sending to processing) - Authorization of transactions based on internal approval mechanism - Deploy Privilege Access Management to counter identity related attacks
Operational	▪ Auto crediting of customer's account based on Pesalink Standard Operating Procedure to credit customers account in less than 60secs. ▪ Auto reversals back into senders account for failed transactions based on Pesalink Standard Operating Procedure. ▪ Real-time transaction monitoring to minimise failure rates (PesaLink Transaction Monitoring System and Transaction Portal can be used as an alternate) ▪ Same Day Reconciliation with Pesalink settlement reports ▪ Prompt dispute resolution and closure on recalls ▪ Prompt action on Fraud Alerts and compliance with internal fraud prevention mechanisms ▪ Prompt customer query resolution through PesaLink helpdesk system or any other channel used by the bank
Anti-Fraud	▪ Setting of transaction limits based on internal risk policies for payment channels, amounts, customer KYC level and any other AML regulations ▪ As a recommendation for fraud mitigation (NOT a MUST) originating bank to consider Multi Factor Authentication (MFA) for transaction originating from channels based on payment value, risk appetite of the bank for transaction limit, KYC and account status to minimise fraud. For example, Transactions through internet banking can use a combination of password AND soft/hardware token while mobile based transactions can only use PIN/Password but with minimal amount and transaction frequency to reduce impact of attempted fraud ▪ Capability for Instant Notification (email or SMS) to customers for all transactions performed in line with best practices for consumer protection